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Advancements and Deployment of Solar Thermal Technologies in Indian Climatic Conditions: A Concise Review

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***Abstract*—Solar thermal technologies are gaining attention as dependable options for high-temperature heat and renewable power generation. This paper reviews the key developments and deployment status of solar thermal and concentrating solar power (CSP) systems under Indian climatic conditions. Recent improvements in absorber coatings, mirror materials, and heat-transfer fluids have increased optical efficiency and system durability, even in dusty and humid environments. The study also examines the role of thermal energy storage (TES), showing that six-to-ten-hour storage can raise plant capacity factors by nearly 25–40% compared with systems without storage. Field data from Indian pilot plants are analyzed to identify common challenges such as mirror soiling, corrosion, and high water consumption. In addition, existing policy gaps are discussed, including limited support for TES and hybridized plants. The paper suggests strategies to better align CSP with India's renewable energy roadmap. Overall, continued innovation in materials, thermal storage, and operation-and-maintenance practices is essential to make solar thermal technologies a viable contributor to India's clean-energy and industrial-heat sectors.**

Keywords*— *Solar Thermal Energy, Concentrating Solar Power (CSP), Thermal Energy Storage (TES), Parabolic Trough, Linear Fresnel Reflector, Solar Tower, Indian Climatic Conditions, Renewable Energy Transition

I. Introduction

Solar thermal technology harnesses the sun's abundant energy to generate usable heat and power, offering a sustainable response to the twin global challenges of rising energy demand and climate change. Unlike conventional photovoltaic (PV) systems that convert sunlight directly into electricity, solar-thermal technologies employ optical and thermal processes to capture and convert solar radiation into heat, serving a wide range of applications from residential heating to industrial-scale power generation [1]. With global energy demand expected to rise by nearly 50 % by 2050 [2], the need for efficient, clean, and dispatchable energy systems has become increasingly urgent.

Concentrating Solar Power (CSP) has emerged as a key option for decarbonizing the energy sector, which currently accounts for about 70% of total global greenhouse-gas emissions [3]. By concentrating and storing solar energy as heat, CSP enables firm and flexible power generation. For instance, a 100-MW CSP plant equipped with six hours of thermal storage can double its operational window, continuing to supply electricity well after sunset [4]. Furthermore, a 5-MW CSP system can prevent nearly 7,000 tons of CO_2 each year—comparable to removing more than 1,500 passenger vehicles from the road [5-6].

In India, renewable-energy deployment continues to accelerate. By 2024, the nation is expected to add nearly 19 GW of new capacity, roughly 85% from solar and 15% from wind sources, while an additional 5–6 GW of coal-based plants remains under development. Despite strong renewable growth, the Ministry of Power projects that around 80 GW of new thermal capacity will still be required by 2031–2032 to meet rising demand [7]. Projections also indicate a likely increase in energy-related carbon emissions, underscoring the urgency of mitigation policies such as Production-Linked Incentive (PLI) schemes for domestic PV manufacturing and greater public-sector procurement of renewable power [8].

To strengthen this transition, India aims to achieve 500 GW of renewable capacity by 2030 [9], reduce overall carbon emissions by one billion tons, and cut carbon intensity by 45%. These national targets align with the country's broader vision of achieving net-zero emissions by 2070 [10]. The growing policy focus on renewables—combined with international momentum toward decarbonization—has spurred rapid technological progress in solar-thermal systems. Recent developments have improved performance, adaptability, and cost-effectiveness, enhancing the practicality of CSP as a major renewable option [11–14].

This review highlights recent advancements in absorber materials, concentrating optics, and thermal-energy-storage

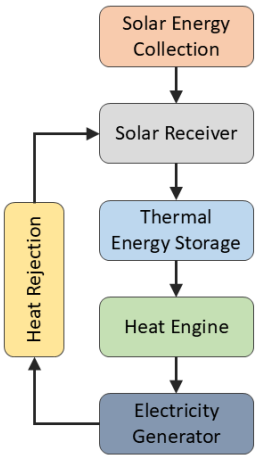

Fig. 1. *Schematic representation of a solar thermal power plant illustrating the conversion of solar radiation into thermal energy with thermal storage*

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(TES) systems, and examines how these innovations address India's specific climatic and operational challenges. It further discusses opportunities for hybridization and auxiliary integration, aiming to support India's renewable-energy roadmap and global decarbonization goals.

II. Fundamentals of Solar Thermal Systems

Solar-thermal technology converts the sun's radiant energy into usable heat that can be applied for electricity generation, water heating, and industrial operations. A solar collector absorbs sunlight through a dark surface and transfers the heat to a circulating working fluid enclosed beneath a transparent glazing. The heated fluid may then be used directly, for example, in solar water-heating systems or indirectly for steam generation in concentrating solar-power (CSP) plants [15–17].

The overall efficiency of a solar-thermal system depends on several key factors, including the collector design, the thermophysical properties of the heat-transfer fluid, and the quality of insulation that minimizes thermal losses. Solar radiation is absorbed by a collector and converted into thermal energy, which is then stored in a thermal-energy-storage system. The stored heat is used to produce superheated steam that drives a turbine–generator to generate electricity. The steam is subsequently condensed and recirculated, allowing continuous power generation, as illustrated in Fig. 1.

A. Types of Solar Collectors

Solar collectors are broadly classified into two categories according to their design and temperature range: flat-plate collectors and concentrating collectors.

1) *Flat-Plate Collectors (FPCs)*
   As shown in Fig. 2, FPCs consist of a dark absorber plate covered by transparent glazing. They perform efficiently at operating temperatures below 100 °C, making them suitable for residential water and space-heating applications [18].
2) *Concentrating Collectors*
   Concentrating collectors employ optical elements, mirrors or lenses, to focus solar radiation onto a smaller receiver area, achieving higher temperatures than flat-plate designs. Their major subtypes include:
   a) *Parabolic-Trough Collectors (PTCs)*
      These use a curved mirror to concentrate sunlight onto a receiver tube along the focal line, producing temperatures up to 400 °C for industrial heat and power generation [19] (Fig. 3).
   b) *Parabolic-Dish Collectors*
      A large, dish-shaped reflector concentrates sunlight onto a focal-point receiver, reaching temperatures of about 800 °C; these systems are generally used for small-scale power production [20] (Fig. 4).
   c) *Linear-Fresnel Reflectors (LFRs)*
      As shown in Fig. 5, a linear array of flat mirrors focuses sunlight onto an elevated receiver, achieving temperatures of approximately 400 °C [21]. This configuration offers a cost-effective alternative to parabolic-trough systems.

The selection of collector type depends on the required temperature level, available space, cost constraints, and intended application.

B. Mathematical Representation

Solar Radiation Intensity (I): It denotes the amount of solar energy incident per unit area during a specific period. It varies with geographical location, time of day, and atmospheric conditions:

$$I = G_{sr}\left(1 - A_f\right)T_f \tag{1}$$

Where I is the solar radiation intensity $\left(\frac{W}{m^2}\right)$, G_{sr} is extraterrestrial solar radiation (constant value, approximately $\frac{1361\,W}{m^2}$, A_f is the atmospheric attenuation factor ($0\ to\ 1$, dependent on weather), and T_f represents the transmittance of the glazing (fraction of solar energy transmitted).

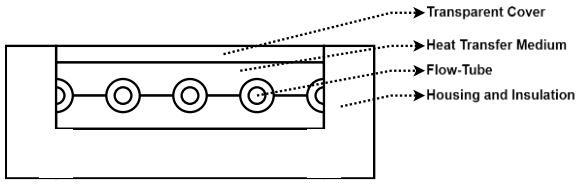

Fig. 2. *Flat plate collectors consist of a dark absorber plate and transparent glazing, suitable for temperatures below 100°C.*

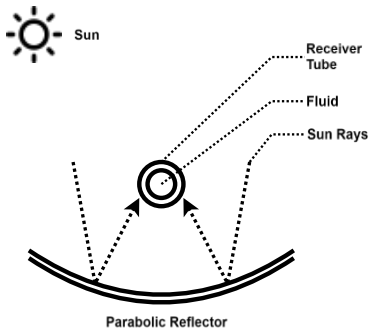

Fig. 3. *A parabolic trough collector utilizes a curved mirror to concentrate sunlight over a receiver tube positioned along its focal line, achieving temperatures of up to 400°C.*

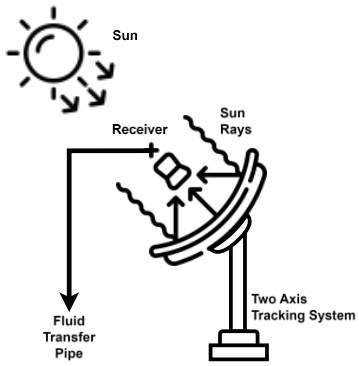

Fig. 4. *A solar dish collector focuses sunlight onto a receiver positioned at its focal point, achieving temperatures of up to 800°C.*

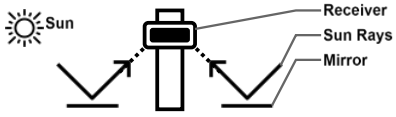

Fig. 5. *Fresnel collectors with a linear array of flat mirrors focusing sunlight onto a linear receiver, achieving temperatures up to 400°C.*

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Solar Collector Efficiency (η_c): It quantifies how effectively incident solar radiation is converted into useful heat transferred to the working fluid:

$$\eta_c = C_R\left(\tau\alpha - U_L\left(T_f - T_a\right)/I\right) \quad (2)$$

Where C_R represents the heat removal factor of collectors, which takes into consideration the heat loss from the collectors, $\tau\alpha$ is the multiplication of the glazing's transmittance (τ) and the absorber plate's absorptance (α), U_L is the overall heat loss coefficient of the collector $\left(\frac{W}{m^2K}\right)$, T_f is the average temperature of the collector fluid (℃), and T_a is the ambient air temperature (℃).

Heat Transfer (H): The useful heat-transfer rate from the collector to the working fluid is determined by

$$H = \eta_C\, A_c\, I \quad (3)$$

Where H is the rate of heat transfer (W), η_C is the efficiency of collector, A_c is the area of collector (m^2), and I is the intensity of solar radiation $\left(\frac{W}{m^2}\right)$.

These expressions describe the fundamental thermodynamic behavior of solar-thermal systems [22]. For the units, W stands for watt, m stands for meter and k is kelvin. The actual parameter values depend on collector type, operating conditions, and location.

III. Overview of Technical and Operational Parameters

CSP plants employ several technologies, Linear Fresnel, Parabolic-Trough, Solar-Tower, and Parabolic-Dish systems, to convert concentrated sunlight into electricity. Their design and performance characteristics differ substantially. Table I summarizes the major technical and operational parameters, including efficiency, land requirement, cost, and working fluids, across the four CSP configurations.

TABLE I. Technical and Operational Parameters Across Different CSPs

S.No.	Ref.	Parameters	Linear Fresnel	Parabolic Trough	Solar Power Tower	Parabolic Disc
1	[23-25]	Cost ($/kW)	2200	4156	4500	6000
2	[26]	Modular Design	Can be added or removed.	Stable structure	Can be easily added	Design can be rotated
3	[23, 26]	Land Footprint	Require less land	Low land required	High	High
4	[23, 27-29]	Scalability	Medium scale, modular	Proven utility-scale (e.g. 50 MW)	Suited for large-scale (>100 MW)	Small-scale modular (5–25 kW units)
5	[30-31]	Environmental Impact	31 g-CO2/kWh	26 g CO2-eq/kWh	38 g CO2-eq/kWh	No
6	[27-28, 32-33]	Efficiency	25-35	30-40	35-45	40-45
7	[34]	Water Usage (L/MWh)	3000-3500	4700	1800-3000	16
8	[35]	Heat transfer fluid	Steam	Synthetic oil	Molten salt	N/A
9	[23]	Cleaning	Regularly needed	Needed	Needed	Needed
10	[32, 36]	Temperature (°C)	200-400	200-400	250-1000	500-1500
11	[37]	Concentration Ratio (W/m^2)	25-60	30-100	500-5000	1000-10000
12	[27, 29, 33]	Power Production scale	Early adoption, industry use	Most deployed (~150 MW)	Limited pilots (e.g. 14 MW)	R&D stage, niche uses
13	[37]	Energy Storage	Possible Water +Steam	Possible Water+Oil+Liquid+Steam	Possible in the form of Heat or molten salt	Possible Stirling Engine
14	[24]	Flexibility	Not Flexible	Relatively flexible	Yes, can work with other grids It has high	Less flexible
15	[23]	Reliability	Low	High	High	High
16	[35]	Hybrid operation	Possible	Possible	Possible	Still in R&D phase
17	[38]	Focus	Line stationery	Line moving	Point stationary	Point stationery
18	[39]	Annual Capacity Factor (%)	22-24	25-28 (no TES) 29-43 (with TES)	55 (with 10h TES)	25-28
Author's Analysis						
19	--	Direct Sunlight	Required	Required	Required	Required
20	--	Maintenance	Simple	Low	High	High
21	--	Greenhouse gasses production	No	Less	No	No
22	--	Life span	10 to 20 yrs	20 to 30 yrs.	Long up to 30 yrs	20 to 30 yrs
23	--	Shape	Linear	Parabolic	massive parabolic	Parabolic shape
24	--	Heat transfer to boiler	Water+Steam	Oil, liquid salt	Air-liquid salt	Water, oil

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TABLE II. Factors Affecting Performance in Solar Thermal Generation [40-44]

Factor	Description	Impact on Performance
Concentration Ratio (C)	The ratio of the absorber plate area to the collector's aperture area.	Flat Plate (C=1): Efficient for low-moderate temperatures. Concentrating (C>1): Achieves higher temperatures but is more complex.
Tracking System	Mechanism that adjusts collector orientation to maintain a direct angle of incidence with the sun.	Fixed: Simpler and cheaper, but efficiency decreases as the sun's position changes. Tracking: Maximizes captured sunlight and improves efficiency (increases cost and complexity).
Solar Radiation	The intensity of sunlight reaching the collector.	Higher radiation leads to higher heat output.
Collector Tilt and Orientation	Angle and direction the collector faces.	Optimal tilt and orientation depend on location and application. Proper positioning increases captured sunlight.
Heat Losses	Transfer of heat from the collector to the surroundings.	Minimizing heat losses through glazing and insulation is crucial for high efficiency.
Flow Rate and Fluid Properties	Rate and thermal properties (e.g., specific heat capacity) of the heat transfer fluid.	Flow Rate: Affects heat absorption and transfer. Fluid Properties: Influence how efficiently heat is carried.

Discussion: As observed, parabolic-trough and tower systems exhibit higher conversion efficiencies (35–45%) and larger scalability (>100 MW), whereas Fresnel systems are cost-effective and require less land. Dish systems, though capable of achieving temperatures up to 800–1500 °C, are generally limited to small-scale or research applications. Factors such as water usage, O&M simplicity, and hybridization potential also influence system selection.

IV. Performance Parameters in Solar-Thermal Generation

The performance of solar-thermal systems depends on multiple environmental and design factors, summarized in Table II. Understanding these parameters is essential for designing efficient collectors and CSP plants suited to India's varied climatic conditions.

V. Advancements in Solar Thermal Technology

The efficiency and temperature outputs of solar thermal collecting systems have been considerably improved by developments in solar thermal technologies. These developments which have been crucial in the expansion of the solar thermal market have been in solar selective coatings, design enhancements, and materials. Furthermore, confirming the continuous development in this field are focused improvements in solar thermal applications on solar collectors and thermal energy storage systems. Many applications of solar energy have benefited from increased efficiency, scalability, and cost-effectiveness attributable in substantial measure to solar thermal technology. These developments are summarized below:

A. Improved Concentrating Solar Power (CSP) Systems

For the purpose of producing electricity, concentrating solar power systems greatly increase the thermal energy by focusing sunlight onto a receiver using mirrors or lenses. More effective solar collector designs include parabolic troughs, linear Fresnel reflectors, and solar power towers define the advances in CSP technology. Furthermore, the integration of advanced tracking systems and control algorithms has optimized the capture of sunlight, leading to improved overall

TABLE III. Comparison between Traditional and Advanced CSP Systems [48-51]

Aspect	Traditional CSP Systems	Advanced CSP Systems
Technology	Typically based on parabolic trough or solar power tower designs	Includes next-generation technologies like advanced parabolic troughs, linear Fresnel reflectors, and solar dish systems
Efficiency	Moderate efficiency levels	Improved efficiency due to technological advancements and innovations
Cost	Initial construction costs are relatively high.	Continued efforts to reduce costs through innovations in materials, manufacturing, and construction techniques
Scalability	Limited scalability due to large-scale infrastructure requirements	Emphasis on modular and scalable designs to accommodate different project sizes and locations
Land Use	Land-intensive, requiring large areas for solar collectors	Efforts to minimize land use through more compact designs and innovative mounting solutions
Thermal Storage	Limited thermal storage capacity, primarily relying on direct electricity generation.	Increasing focus on integrating thermal energy storage for enhanced dispatch ability and grid integration
Reliability	Well-established technology with proven reliability	Ongoing improvements to enhance system reliability and performance under varying weather conditions
Flexibility	Limited flexibility in system configuration and operation	Increasing flexibility through modular designs and advanced control systems, allowing for optimized performance and grid integration
Environmental Impact	Generally, there is a low environmental impact compared to fossil fuels, but land use can be a concern.	Efforts to minimize environmental impact through improved land use practices and technological innovations
Market Adoption	It is widely deployed in utility-scale projects, particularly in regions with high solar irradiance.	Increasing adoption globally, driven by declining costs, supportive policies, and growing recognition of CSP's role in decarbonizing the energy sector

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system performance [45-47]. A comparison of traditional and new CSP systems is presented in Table III.

B. Enhanced Solar Collectors

Innovations in solar collector design have increased the efficiency of capturing solar energy for thermal and electrical applications. Evacuated tube, flat plate, and concentrating collectors have been refined to maximize energy absorption and minimize heat loss. The use of selective coatings and high thermal conductivity materials has improved the performance of solar collectors in various climatic conditions. Efficient heat removal from solar collectors is crucial for maximizing performance in solar thermal systems. Multiple techniques are used to enhance the effectiveness of collectors, which may be categorized into three main groups:

Increasing Heat Transfer Coefficient: One method is to enhance the heat transfer coefficient between the absorber plate or tube and the working fluid [52]. A method to do this is by using artificially roughened absorber surfaces, which augment the effective surface area for heat transmission and improve thermal performance.

Special Coatings: An alternative method involves using specialized coatings on the surface of the absorber. Solar-selective coatings are designed to absorb solar radiation while selectively minimizing heat loss through emission. These coatings typically consist of multiple layers optimized for high sunlight absorptance and low infrared radiation emissivity, thereby maximizing thermal efficiency [53].

Nanofluids: It consists of nanoparticles diffused in a base fluid, offering another way to enhance heat conductivity. The total thermal conductivity of the working fluid is improved by adding highly thermal conductive nanoparticles into it, hence increasing the heat transfer efficiency in the solar collector [54]. To enhance the thermal performance of CSP systems, nanofluid-based heat transfer media and advanced selective coatings are increasingly being adopted. Nanofluids, such as Al_2O_3, CuO, SiO_2 [55, 57] and carbon-based nanoparticles dispersed in synthetic oils or molten salts, can increase the thermal conductivity by 10–30% [55], leading to a 3–7% improvement in overall system efficiency. Similarly, spectrally selective coatings applied to receiver tubes can yield solar absorptance of 0.95–0.97 and emittance below 0.15, significantly minimizing radiative losses [56].

However, both these technologies face practical limitations that affect their commercial scalability. Nanofluids exhibit issues such as particle agglomeration, sedimentation, and erosion of receiver surfaces, which can degrade heat transfer efficiency over time and increase maintenance costs. The synthesis of stable nanofluids also adds 10–20% to the overall heat transfer fluid (HTF) cost [57]. Similarly, advanced ceramic–metal (cermet) coatings and multi-layer selective absorbers, though highly efficient, suffer from thermal degradation and oxidation under prolonged exposure at temperatures above 550°C, reducing operational life and necessitating periodic recoating, which raises maintenance and manufacturing costs by 15–25% [57-58].

Numerous studies have investigated these methods and their impact on collector performance. Research in this field focuses on developing innovative approaches to enhance heat removal from solar collectors, thereby improving the overall efficiency and effectiveness of solar thermal systems, which has become a key focus area in recent developments. Table IV provides a comparison between conventional and advanced technologies in this domain.

C. Advanced Thermal Energy Storage (TES) Systems

Enabling continuous power production and reducing the irregularity of solar energy depend on thermal energy storing. High-temperature phase change materials (PCMs), creative approaches for transforming thermal energy into electrical energy, the use of molten salt and advanced ceramics to enable effective storage and release of heat, all of which constitute recent developments in thermal energy storage (TES) systems. By means of the integration of TES systems with CSP plants, dispatchable power production has been made possible, hence improving grid reliability and stability [62–64]. Table V provides an expansive summary of various advanced TES systems.

D. Innovative Materials and Components

Scientists have concentrated their efforts on creating sophisticated materials and components to improve the durability, reliability, and efficiency of solar thermal systems. High-performance absorber coatings, such as selective surfaces and nanomaterials, have been engineered to increase solar absorption and reduce thermal emittance. Advances in mirror materials, including glass-metal composites and reflective polymers, have improved reflectivity and resistance to degradation [71-72].

TABLE IV. COMPARISON BETWEEN CONVENTIONAL AND ADVANCED TECHNOLOGIES OF ENHANCED SOLAR COLLECTOR [58-61]

Metric	Conventional Technologies (Flat Plate Collectors)	Advanced Technologies
Design	Simple, non-concentrating	Concentrating (parabolic trough, dish) or advanced flat plate designs
Concentration Ratio	Low (no concentration)	High (concentrates sunlight)
Operating Temperature	Low to moderate (up to 100°C)	High (can reach several hundred °C)
Efficiency	Moderate (typically 60-80%)	Higher potential efficiency (up to 90%)
Applications	Domestic hot water, low-temperature space heating	High-temperature process heat, power generation, desalination
Materials	Copper tubes, single glazing	Selective coatings, advanced absorber materials, evacuated tubes (for some designs)
Durability	Proven track record	Newer technology's long-term durability needs evaluation

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TABLE V. COMPARISON BETWEEN CONVENTIONAL AND ADVANCED TES SYSTEM TECHNOLOGIES [65-70]

Metric	Conventional Technologies (Sensible)	Advanced Technologies (Latent/Thermochemical)
Storage Mechanism	Heat absorbed/released via material temperature rise/drop (e.g., molten salts at 300-600°C in CSP). Simple, but volume-intensive.	Phase transitions (latent: melting/freezing) or reversible reactions (thermochemical: adsorption/desorption). Enables isothermal storage, minimizing losses.
Material	Abundant/low-cost (water: ~4.18 kJ/kg·K; rocks: ~0.8-1.0 kJ/kg·K).	Advanced (PCMs: paraffin/salt hydrates at 150-300 kJ/kg; zeolites: >500 kJ/kg via sorption). Tailorable melting points.
Operating Range	Typically, <200°C (e.g., water/rocks); low flexibility for high-heat industries.	0-1000°C+; PCMs customized (e.g., 18-28°C for buildings).
Energy Density	50-200 kWh/m³ (water lowest at ~0.1 kWh/kg).	100-500+ kWh/m³ (PCMs 2-3x sensible; thermochemical up to 10x).
Maturity	Commercial (e.g., Godawari CSP plant, Rajasthan: 50 MW molten salt).	Pilot/demo (e.g., IIT Bombay CO2-TES R&D); global CSP pipeline includes India.
Applications	Short-term grid support, district systems.	Process heat, HVAC in buildings; cold chain (e.g., PCM vaccines in India).
Future potential	Cost reductions via scaling (target: <INR 50 crore/MW by 2030).	Encapsulation advances for longevity; hybrid with batteries.
India-Specific Regulatory & Incentive Framework	Well-aligned with existing IEGC dispatch codes for scheduling; benefits from ECBC for buildings and general energy efficiency incentives (e.g., accelerated depreciation, GST exemptions); limited unique barriers but challenged by grid intermittency.	Lacking specific dispatch recognition in IEGC/CERC regs (e.g., no dedicated ancillary services for TES); incentives skewed to batteries (VGF up to 40%, proposed PBI INR 1-2/kWh); high costs and awareness gaps hinder adoption—pilots needed for solar thermal/CSP integration.

E. Hybridization and Integration

Hybridizing solar thermal systems with other renewable energy technologies, such as photovoltaics and biomass, can increase energy production and system efficiency. Integrating solar thermal systems with conventional power plants, district heating networks, and industrial processes offers opportunities for combined heat and power generation and process heat applications [73].

CSP plants in India face unique climate-related challenges, with dust and humidity having a particularly pronounced impact on plant performance and operational reliability. Dust accumulation, especially in arid regions like Rajasthan and Gujarat, can lead to efficiency losses of up to 25%, as particulates block sunlight from reaching receiver surfaces and mirrors. Frequent dust storms and dry winds further exacerbate the issue, making regular cleaning and maintenance essential for consistent power generation and long-term durability. Coastal zones and areas exposed to monsoon conditions add another layer of complexity, as high humidity and rainfall can cause corrosion of components, algal growth, and rapid degradation of advanced coatings, necessitating specialized cleaning and mitigation strategies to maintain efficiency [74-75].

To address India's fluctuating direct normal irradiance and intermittency problem, several hybridization pilot projects have been launched, mainly combining CSP with coal or photovoltaics. For instance, the Dadri coal plant hybridization in Uttar Pradesh integrates a CSP solar field with an existing steam turbine, intended to lower startup times and boost output. Similar hybrid pilots are in various stages of planning or deployment in states like Tamil Nadu, Andhra Pradesh, and Chhattisgarh, where the proximity of solar resources to existing coal infrastructure presents opportunities to enhance capacity factors and operational flexibility. These projects demonstrate a pragmatic approach to India's energy transition, leveraging CSP hybridization to mitigate climatic and technical limitations and improve renewable integration within the grid [76-77].

F. Digitalization and Control Systems

Implementing digitalization and intelligent control systems enables real-time monitoring, predictive maintenance, and optimization of solar thermal plants, maximizing energy yield and operational efficiency. Combining cutting-edge control algorithms, machine learning, and artificial intelligence (AI) approaches helps solar thermal systems to be adaptable and autonomous. These latest developments help solar thermal technology to grow and mature by releasing its possibilities to be a reliable and sustainable energy source for numerous applications [78], including electricity generation, industrial process heat, water desalination, and space heating and cooling. Maintaining constant innovation and cooperation across academic, business, and government sectors will help to hasten the

TABLE VI. SOLAR THERMAL TECHNO-ECONOMIC ANALYSIS [81-87]

Factor	Description	Data (India Specific)	Units
Capital Costs	Initial investment for building the plant	₹15 Cr - ₹30 Cr	Crore INR/MW
O&M Costs	Ongoing maintenance expenses	₹50 Lakh - ₹1 Cr	Lakh INR/MW per year
Fuel Costs	No fuel is required for electricity generation	₹0	INR/kWh
Energy Efficiency	Percentage of solar radiation converted to electricity	18% - 22%	%
Capacity Factor	The ratio of average power output to plant capacity	45% - 50%	%
Levelized Cost of Energy	Average lifetime cost of electricity generation	₹5.5 - ₹8.0	INR/kWh

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TABLE VII. GLOBAL LCOE AND CAPITAL COST COMPARISON OF RENEWABLE TECHNOLOGIES (2010–2024) [88-90]

Technology	LCOE 2010 (US$/kWh)	LCOE 2024 (US$/kWh)	LCOE decline (2010→2024)	TIC (Installed cost) 2010 (US$/kW)	TIC 2024 (US$/kW)	TIC decline (2010→2024)	CapEx rel (PV=1.00)	CapEx rel (max=1.00)	Discussion (Author)
Solar PV (utility-scale)	0.417	0.043	89.69%	5,283	691	86.92%	1.00	0.188	Biggest absolute LCOE and TIC fall; now one of the cheapest new resources.
Wind (onshore)	0.113	0.034	69.91%	2,324	1,041	55.21%	1.51	0.283	Lowest LCOE in 2024 (slightly cheaper than PV); moderate TIC decline.
CSP (with storage)	0.402	0.092	77.11%	10,703	3,677	65.65%	5.32	1.00	Still highest-capex and highest LCOE of the three, but large improvements since 2010, especially where longer storage is deployed.

worldwide adoption and deployment of solar thermal technology. In conclusion, research in solar thermal technology for water evaporation and clean water production has significantly progressed in recent years. By leveraging advancements in materials science, chemistry, and membrane technologies, researchers are developing innovative solutions [79] to harness solar energy for sustainable water purification applications efficiently.

VI. TECHNO-ECONOMIC ANALYSIS AND APPLICATIONS

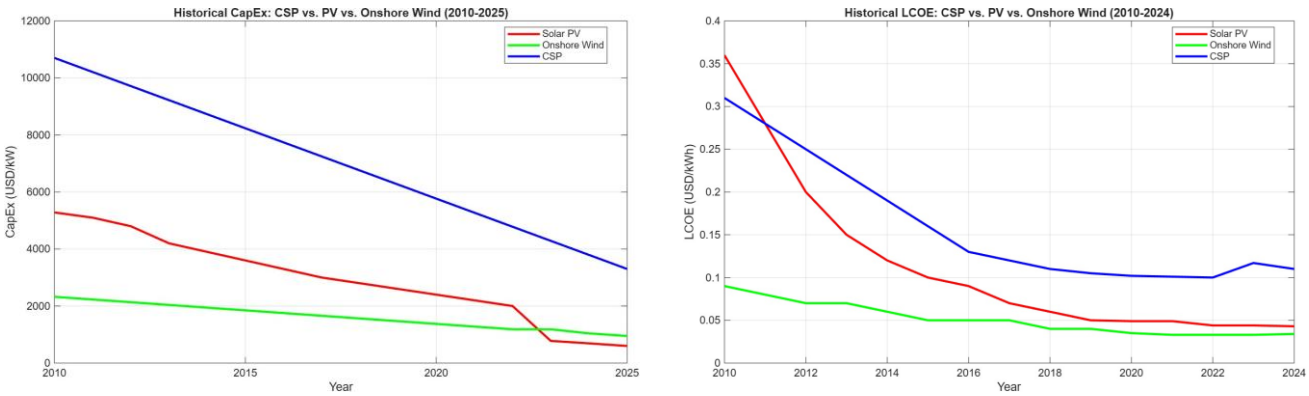
Ongoing research and development continue to reduce the capital cost and *Levelized Cost of Electricity (LCOE)* of solar-thermal systems through technological innovation and economies of scale [80]. Recent improvements in collector geometry, receiver coatings, and high-temperature heat-transfer fluids have enhanced system efficiency and durability. Integration of *Thermal Energy Storage (TES)* enables on-demand power generation and improves grid reliability, although the additional cost must be weighed against operational benefits. As solar-thermal technology matures, further cost reductions are anticipated, even though complex *Concentrating Solar Power (CSP)* configurations may still involve higher initial investments, as shown in Table VI.

Table VII summarizes global techno-economic trends for major renewable-energy technologies, including solar photovoltaic (PV), onshore wind, and CSP, over the period 2010–2024. PV exhibits the steepest decline in both LCOE and *Total Installed Cost (TIC)* due to large-scale manufacturing and module-price reductions, followed by wind, which benefits from aerodynamic and turbine-design advances. Although CSP remains capital-intensive, its costs have fallen by approximately 77 % in LCOE and 66 % in TIC, driven by improved optical materials, high-temperature receiver design, optimized plant layouts, and TES integration. Fig. 6 illustrates the historical decline in both LCOE and Capital Expenditure (CapEx) for these technologies during 2010–2025, where Fig. 6(a) shows LCOE trajectories and Fig. 6(b) presents CapEx evolution. The combined data confirm the global downward trend in renewable-energy costs and demonstrate the emerging techno-economic viability of CSP with TES as a dispatchable, low-carbon power source.

Discussion: As observed from Tables VI–VII and Fig. 6(a–b), utility-scale PV demonstrates the most rapid cost reduction and scalability, whereas onshore wind shows consistent improvement through aerodynamic and structural enhancements. CSP, though comparatively costlier, provides unique advantages of *dispatchability* and *high-temperature process heat*. TES integration extends generation beyond daylight hours and increases reliability under India's high-irradiance climatic conditions, underscoring CSP's strategic contribution to a balanced and resilient renewable-energy mix.

VII. CASE STUDY

India presently operates a number of solar-thermal power plants that use different *Concentrating Solar Power (CSP)* technologies such as parabolic-trough, solar-tower, linear-Fresnel, and paraboloidal-dish systems. These projects cover a wide range of power capacities and reflect both research-oriented and commercial-scale implementations. Among them, parabolic-trough plants are the most common,



(a) (b)

Fig. 6. Historical cost trends for major renewable-energy technologies, (a) Levelized Cost of Electricity (LCOE) trajectories for utility-scale PV, onshore wind, and CSP from 2010 to 2024, and (b) Capital Expenditure (CapEx) evolution in USD per kW for the same technologies from 2010 to 2025.

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supported by several Indian as well as international manufacturers.

The working fluids generally consist of synthetic thermal oils and molten salts, which are effective in absorbing heat and providing medium- to high-temperature thermal-energy storage. Many plants also include *Thermal Energy Storage (TES)* units to ensure reliable power generation even after sunset. The combination of different technologies, plant capacities, and commissioning years shows India's growing commitment to solar-thermal power as a dependable and environment-friendly energy source. Key operational and demonstration projects, along with their specifications and current status, are summarized in Table VIII.

VIII. Conclusion

Solar-thermal technologies can support India's clean-energy transition by supplying high-temperature heat and dependable electricity. Recent progress in collectors, absorber coatings, heat-transfer fluids, and thermal-energy storage has improved performance and reduced costs. Our review and case studies show that plants designed for Indian conditions can operate reliably when dust control, corrosion protection, and water use are managed carefully. Techno-economic trends also indicate a steady decline in LCOE and capital costs, with TES adding clear value by extending generation beyond daylight hours.

At the same time, important gaps remain. Long-term durability of selective coatings and nanofluids under high temperatures needs further validation. Local manufacturing of mirrors, receiver tubes, and TES components should be scaled to lower project costs. Policy support that recognizes the system value of dispatchable solar heat and power will accelerate adoption, especially for industrial process heat. Future work should focus on hybrid configurations with PV and biomass, standardized O&M for dusty and humid climates, and digital monitoring for AI-based predictive maintenance. With coordinated efforts from academia, industry, and government, solar-thermal technologies can

TABLE VIII. Case Study of Operational or Non Operational Power Plants in India [91-108]

S. No.	Project (place)	Company / manufacturer	Heat transfer fluid	Capacity (MW)	Technology type	Advancement	Year	Current status
1	Abhijeet Solar Project, Phalodi (Jaisalmer, Rajasthan)	Corporate Ispat / Abhijeet (planned)	Therminol VP-1 (planned)	50 (planned)	Parabolic trough (planned)	—	2013–2015 (planned)	Project cancelled / converted / not commissioned
2	ACME Solar Tower, Bikaner (Rajasthan)	ACME Group / eSolar (demo)	Water/steam (direct steam)	2.5 (demo)	Power tower (solar tower)	Demonstrator (no long-duration TES at demo scale)	2011	Operational
3	Dhursar (Jaisalmer, Rajasthan)	Reliance Power / Rajasthan Sun Technique	(CLFR field; HTF / steam)	125	Linear Fresnel (CLFR)	No large TES (field supplies steam / hybrid)	2014	Operational
4	Diwakar (Askandra / Askandra village, Jaisalmer)	Lanco / Diwakar (developer names vary)	Synthetic oil / molten salt (various plans)	100 (planned / contracted)	Parabolic trough (planned)	Claimed: 3–4 hr TES in some docs; implementation unclear	2013 (award)	Status: unclear /
5	Godawari Solar Project, Nokh (Jaisalmer, Rajasthan)	Godawari Green Energy Ltd (GGEL)	Dowtherm A / synthetic oil	50	Parabolic trough	No large TES reported in public entries	2013	Operational
6	Gujarat Solar One (Rapar / Kutch, Gujarat)	Cargo Power / Cargo Solar (owner)	Diphyl / EuroTrough design (hot oil)	25	Parabolic trough	~9 hr TES (reported design)	2014 (commissioned)	Operational
7	KVK Energy Solar (Askandra, Jaisalmer)	KVK Energy Ventures / (ties with Lanco historically)	Synthetic oil	100	Parabolic trough	Claimed 4 hr TES in developer documentation	2013	Non-operational
8	Megha Solar Plant, Anantapur (Andhra Pradesh)	Megha Engineering & Infrastructure	Synthetic oil	50	Parabolic trough		2014	Operational
9	National Solar Thermal Power Facility (Gwalpahari / Gurgaon, Haryana)	IIT Bombay (R&D facility; EPC Abengoa historically)	Therminol VP-1 (hot oil) & hybrid LFR / trough elements	1 (research / demo)	Parabolic trough + LFR (hybrid demo)	Research / test facility (no utility-scale TES requirement)	2012	Operational
10	India One (Abu Road, Rajasthan)	WRST / World Renewal Spiritual Trust (Brahma Kumaris)	Water (dish receivers with iron cores)	1	Parabolic dish (dish-engine / direct steam)	approx16 hr thermal storage (iron receiver cores) reported	2011 / 2017 (various sources cite commissioning)	Operational

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become a stable and scalable part of India's low-carbon energy portfolio.

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